

American Express Data Analyst Interview Questions

SQL Questions

1. Write a query to find duplicate rows in a table.
2. Explain the difference between INNER JOIN and OUTER JOIN with examples.
3. Write a query to fetch the second-highest salary from an employee table.
4. How do you use GROUP BY and HAVING together? Provide an example.
5. Write a query to find employees earning more than their managers.
6. What is a window function in SQL? Provide examples of ROW_NUMBER and RANK.
7. Write a query to fetch the top 3 performing products based on sales.
8. Explain the difference between UNION and UNION ALL.
9. How do you use a CASE statement in SQL? Provide an example.
10. Write a query to calculate the cumulative sum of sales.
11. What is a CTE (Common Table Expression), and how is it used?
12. Write a query to identify customers who have made transactions above \$5,000 multiple times.
13. Explain the difference between DELETE and TRUNCATE commands.
14. How do you optimize SQL queries for better performance?
15. Write a query to find all customers who have not made any purchases in the last 6 months.
16. How do you handle NULL values in SQL? Provide examples.
17. Write a query to transpose rows into columns.
18. Explain indexing and how it improves query performance.
19. Write a query to fetch the maximum transaction amount for each customer.
20. What is a self-join, and how is it used?
21. [Very Hard] Write a query to find consecutive transactions with the same amount for each customer.
22. [Very Hard] Write a query to identify gaps in transaction dates for each customer.
23. [Very Hard] Write a query to rank transactions within each month per customer using window functions.

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- 24. [Very Hard] Write a query to detect the 3-day rolling average of transaction amounts for each user.
- 25. [Very Hard] Write a query to detect the first and last transaction date for each customer in each year.

Data Analysis / Scenario-Based Questions

- 21. How would you design a database to store credit card transaction data?
- 22. Write a query to identify the most profitable regions based on transaction data.
- 23. How would you analyze customer churn using SQL?
- 24. Explain the difference between OLAP and OLTP databases.
- 25. How would you determine the average revenue per user (ARPU) from transaction data?
- 26. Describe a scenario where you would use a LEFT JOIN instead of an INNER JOIN.
- 27. Write a query to calculate YoY (Year-over-Year) growth for a set of transactions.
- 28. How would you implement fraud detection using transactional data?
- 29. Write a query to find customers who have used more than 2 credit cards for transactions in a given month.

Behavioral and Business Questions

- 30. How would you approach a business problem where you need to analyze the spending patterns of premium customers?